

The Config Bench

V9 — Power Law $P = I \times V$

Episode TCB-PWR-001 | John King | @TheConfigBench | Target: ~8–10 min

Episode Overview: Follows directly from V8 (Ohm's Law Questions & Giveaway). Opens with the V8 giveaway winner announcement. Introduces the Power Law ($P = I \times V$) and its three derived forms. Uses Xournal++ for formula walkthrough, Qucs for circuit simulation, and a live bench demo to show power dissipation in a resistor. Closes with 4 test questions, hand-packed component starter pack giveaway announcement, and a preview of V10.

Episode Timing Overview

Seg	Content	Est. Time	Running Total
1	Intro + V8 Giveaway Winner Announcement	1:00	1:00
2	What is Power? Real-world context	1:00	2:00
3	The four Power Law formulas (Xournal++)	1:30	3:30
4	Qucs: Simulate power in a resistor circuit	1:30	5:00
5	Bench demo: Measure V, I, calculate P, feel the heat	2:00	7:00
6	4 Questions + Component Pack Giveaway + V10 preview	1:30	8:30

SEGMENT 1 — Intro + V8 Giveaway Winner Announcement

SEGMENT 1 | Est. ~1:00

■ CAMERA: Bench shot. You on camera, relaxed. Standard intro framing.

Welcome back to The Config Bench — I'm John.

Before we get into today's content, I need to take care of some business from V8.

If you caught the last episode, you know I put out six Ohm's Law questions and told you whoever answers all six correctly with their math shown wins the FNIRSI DSO152 mini oscilloscope.

ACTION: Hold up the oscilloscope briefly on camera.

We had a winner — and fast. Twelve views in and someone came through with every single answer correct, math shown, formulas clean. Exactly what I asked for.

The winner is Lewis Shippy. Congratulations Lewis — your prize is on its way.

That's what learning electronics looks like. Do the work, show the math.

Alright. V8 was Ohm's Law. Today we go one step further. Today is Power.

SEGMENT 2 — What Is Power? Real-World Context

SEGMENT 2 | Est. ~1:00

■ CAMERA: Stay on bench shot. Conversational — no props yet.

You've been using the word 'power' your whole life. 60-watt bulb. 65-watt laptop charger. 750-watt power supply. But what does that number actually mean?

Power is the rate at which energy is consumed or delivered. In electronics, it's measured in Watts — named after James Watt.

And the formula that ties it all together is dead simple:

$$P = I \times V$$

Power equals Current times Voltage. That's it.

A 60-watt bulb on 120 volts draws half an amp. Your 65-watt laptop charger on a 20-volt USB-C port pulls 3.25 amps. Every wattage rating you've ever seen comes from P equals I times V.

Here's why it matters on the bench: every resistor has a power rating. Quarter-watt. Half-watt. Two watt. Push more power through it than it's rated for and it gets hot, fails, and takes your circuit with it. Knowing the power is how you choose the right component before you build.

SEGMENT 3 — The Four Power Law Formulas (Xournal++)

SEGMENT 3 | Est. ~1:30

■ XOURNAL++: Open Xournal++. One page per formula group. Write large, clean.

Okay — Xournal. Let's write this out.

$P = I \text{ times } V$ is the base formula. But just like Ohm's Law gave us three versions by rearranging, Power Law gives us four useful forms when you combine it with Ohm's Law.

Formula	Use it when you know...	Solve for...
$P = I \times V$	Current and Voltage	Power
$P = I^2 \times R$	Current and Resistance	Power (no voltage needed)
$P = V^2 / R$	Voltage and Resistance	Power (no current needed)
$I = P / V$	Power and Voltage	Current draw

■ XOURNAL++: Write each formula on its own line. Circle $P = I \times V$ as the anchor. Draw arrows showing the derivations.

$P = I \text{ times } V$ is your anchor. The others come from substituting Ohm's Law in.

Substitute $V = IR$ and you get $P = I\text{-squared times } R$. Use it when you know current and resistance but haven't measured voltage.

Substitute $I = V \text{ over } R$ and you get $P = V\text{-squared over } R$. Use it when you know voltage across a component and its resistance.

You don't need to memorize all four right now. Just know they exist and where they come from.

SEGMENT 4 — Qucs Simulation: Power in a Resistor Circuit

SEGMENT 4 | Est. ~1:30

■ QUCS: Build: 9V DC source, 100Ω resistor. Run DC simulation. Read V, I, P.

Let's go to Qucs.

Nine volt DC source, one hundred ohm resistor. We already know the current from Ohm's Law — nine divided by one hundred — ninety milliamps.

Run the simulation. Read off voltage and current.

Qucs expected results (9V / 100Ω):

$V = 9V$ | $I = 0.09A$ (90mA) | $P = I \times V = 0.09 \times 9 = \mathbf{0.81W}$

Check: $P = I^2R = 0.09^2 \times 100 = \mathbf{0.81W}$ ✓ | $P = V^2/R = 81/100 = \mathbf{0.81W}$ ✓

Point eight one watts. What does that mean for parts selection?

A standard resistor from your bin is rated quarter watt — 0.25W. This circuit needs 0.81 watts. That resistor will overheat and fail.

You need at minimum a one-watt part. Two-watt for thermal headroom.

Run the calculation before you build. Not after things start smoking.

■ QUCS: Swap to 1000Ω. Same 9V. Power drops to 81mW — quarter-watt resistor is now fine.

Swap to one thousand ohms. Same source.

Current drops to nine milliamps. Power drops to eighty-one milliwatts. Now a quarter-watt resistor has plenty of margin. Same voltage, ten times the resistance, one tenth the power. That's $P = V\text{-squared over } R$ doing exactly what it should.

SEGMENT 5 — Bench Demo: Measure It, Feel It

SEGMENT 5 | Est. ~2:00

■ CAMERA: Tight bench shot. Breadboard, 9V battery, 100Ω resistor, multimeter.

Off the screen and onto the bench.

Nine volt battery. One hundred ohm resistor on the breadboard.

Multimeter across the resistor — measure voltage.

ACTION: Measure and read voltage aloud. Should be ~9V.

Now current. Meter to milliamps, in series.

ACTION: Measure and read current aloud. Should be ~90mA.

$P = I \text{ times } V$. Ninety milliamps times nine volts. Point eight one watts.

Now — touch the resistor.

ACTION: Touch the 100Ω resistor. It will be warm. Comment on it.

Feel that? That's 0.81 watts being turned into heat. That's what power dissipation means — the energy goes somewhere, and in a resistor it goes into heat.

Imagine that part is only rated for a quarter watt. At four times its rating it doesn't just get warm — it gets hot enough to fail, discolor, and smoke. Always check your power math before you build.

SEGMENT 6 — Test Questions, Component Pack Giveaway + V10 Preview

SEGMENT 6 | Est. ~1:30

■ CAMERA: Standard bench framing. Xournal++ for questions.

Four questions. Same rules. Show your math — no math, no entry.

Test Questions — V9

Q1: A circuit has 12V across a 60Ω resistor. What is the power dissipated?

- A) 0.2W
- B) 2.4W
- C) 2.4mW
- D) 720W

Formula: $P = V^2 / R = 12^2 / 60 = 144 / 60 = 2.4W$

Answer: B) 2.4W

Q2: A current of 500mA flows through a 10Ω resistor. What is the power?

- A) 5W
- B) 2.5W
- C) 0.25W
- D) 50W

Formula: $P = I^2 \times R = (0.5)^2 \times 10 = 0.25 \times 10 = 2.5W$

Answer: B) 2.5W

Q3: A 9V battery powers a device drawing 300mA. What is the power consumption?

- A) 30W
- B) 27W
- C) 2.7W
- D) 0.033W

Formula: $P = I \times V = 0.3A \times 9V = 2.7W$

Answer: C) 2.7W

Q4: You have a 1/4W resistor rated at 100Ω. What is the maximum safe voltage across it?

- A) 25V
- B) 5V
- C) 0.05V
- D) 2.5V

Formula: $V = \sqrt{(P \times R)} = \sqrt{(0.25 \times 100)} = \sqrt{25} = 5V$

Answer: B) 5V

GIVEAWAY ANNOUNCEMENT — Hand-Packed Component Starter Pack

■ **CAMERA:** *Hold up a small zip bag or component pouch on camera — doesn't need to be the final packed bag, just something visual.*

■ **PRIZE:** Hand-curated electronics component starter pack A personally selected mix of components to get you building — resistors, capacitors, diodes, transistors, jumper wires, and more. Plus wire and alligator clips so you can connect a battery to your breadboard and start experimenting from day one. Exact contents vary — every pack is put together by hand.

For V9 — the prize is a component starter pack.

I'm putting these together by hand — a mix of resistors, capacitors, diodes, transistors, jumper wires, some hookup wire, and alligator clips so you can connect a battery straight to your breadboard without needing anything else.

Everything you need to start building real circuits on a real breadboard. I'm not going to tell you exactly what's in it — that's part of the fun. But I can tell you there's enough in there to keep you busy.

Answer all four questions with your math shown, drop it in the comments. One winner gets the pack shipped to their door.

Subscribe — winner announced at the top of V10.

CLOSING

That's Power Law. $P = I \text{ times } V$ — and now you've got all four forms in your toolkit.

Ohm's Law gives you voltage, current, and resistance. Power Law tells you what the circuit is doing with energy. Together those two cover the foundation of everything you'll build on this bench.

V10 is series and parallel resistor networks — we start putting these formulas to work in real multi-component circuits. That's where it gets interesting.

I'm John. This is The Config Bench. See you in V10.

YOUTUBE METADATA — V9

Title

Power Law Explained: $P = I \times V$ | The Config Bench V9

Description

V8 giveaway winner announced at the top — congratulations Lewis!

In V9 we cover the Power Law — $P = I \times V$ — and all four derived forms. Live bench demo shows power dissipation in a 100Ω resistor and why choosing the right wattage rating matters before you build. Qucs simulation confirms the math.

Four questions at the end — show your math to enter the V9 giveaway: a hand-packed electronics component starter pack with everything you need to start building real circuits on a breadboard.

- Complete Electronics Self-Teaching Guide (Boysen & Kybett): [\[LINK\]](#)
- Subscribe for V10 — Series & Parallel Resistor Networks

#electronics #powerlaw #TheConfigBench #ohmslaw #watts #diyelectronics #learnelectronics #giveaway

Tags

power law, $P=IV$, watts law, electronics for beginners, bench electronics, resistor wattage, power dissipation, ohms law, qucs simulation, TheConfigBench, learn electronics, DIY electronics, circuit basics, electronics giveaway, component kit, V9

Pinned Comment

4 questions are in the video — drop ALL your answers in the comments with your math shown. No math = no entry. One subscriber who answers all four correctly wins a hand-packed electronics component starter pack. Winner announced in V10. Subscribe so you don't miss it! ■